

Data Mining Complete Exam Notes

CT 72502 (Elective I) · BEI / BEX IV-I · TU, IOE · tiers over 23 papers, 2069–2082 BS

6. Anomaly / Fraud Detection

3 hrs · in 23/23 papers, the only chapter with perfect coverage · syllabus gives it no sub-topics · 1 numerical → Ch 6 companion.

Read this first:

- Only 3 hrs of syllabus, but it has appeared in **every single paper**. **Best marks-per-hour in the subject**.
- One question dominates: **“What is anomaly detection + explain distance-based methods”** **TOP 10/23**. Learn that and the importance/types answer, and the chapter is banked.

6.1 Basics

What is anomaly detection? Why is it important? 2+2+4 3+3 5 2+3

HOT 6/23 82 Bh, 81 Bh, 81 Ba, 79 Bh, 73 Ch, 72 Ka

Definition

- An **outlier / anomaly** is a data object that **deviates significantly** from the rest, as if generated by a **different mechanism**.
- **Anomaly detection** = finding those objects.
- Hawkins: an observation deviating so much that it arouses suspicion it was generated by a different process.

Outlier ≠ noise

- **Noise** = random error / variance in a measured value. Has **no meaning**, should be **removed** before mining.
- **Outlier** = a genuine object from a different mechanism. Carries **information**, and is often *the thing you are looking for*.
- **Same data point, opposite treatment** depending on the task.
- Related: **novelty detection** finds genuinely new patterns; once confirmed they are *folded into* the normal model, so later instances are no longer flagged.

Why it matters / applications

- **Fraud detection**: credit-card use unlike the customer's normal pattern.
- **Intrusion detection**: traffic unlike normal network behaviour.
- **Medical**: unusual symptoms or test results → disease.
- **Public health**: a spike in one region → outbreak.
- **Industry**: fault and defect detection, equipment failure.
- **Ecosystem disturbance**: hurricanes, floods, droughts.
- **Data cleaning**: find and remove genuinely erroneous values.

The one-line justification

- In these applications the **rare event is the valuable event**. Ordinary mining is tuned to find *frequent* patterns, so it actively discards exactly what matters here.

Types of outliers (80 Bh asks this directly, 5)

- **Global outlier (point anomaly)**
 - Deviates significantly from **the rest of the whole data set**.
 - Simplest type; most methods target these.
 - Eg a computer broadcasting a huge number of packets in a short time.
- **Contextual (conditional) outlier**
 - Deviates significantly **with respect to a chosen context**.
 - Attributes split into two groups:
 - o **Contextual attributes**: define the context. Eg date, location.
 - o **Behavioural attributes**: what is judged. Eg temperature, humidity.
 - Eg **28°C is an outlier in a Kathmandu winter, normal in summer**.
 - Generalises **local outliers**; a global outlier is the special case where the context is empty.
- **Collective outlier**
 - A **subset of objects that as a whole** deviates, even though **no individual member is an outlier**.
 - Eg 1 delayed shipment is normal; **100 delayed on one day** is a collective outlier.
 - Eg a burst of packets from many machines at once = denial-of-service, though each single packet looks fine.
 - Detecting them needs the **relationships** between objects (sequence, spatial, graph structure), not just the objects.

Challenges / issues 76 Ch 2, 72 Ch short note

- Defining “normal” is hard, and the boundary is rarely sharp.
- Normal behaviour **evolves** → today’s model is stale tomorrow.
- Application-specific: one definition does not transfer between domains.
- **Scarcity of labelled data** for training and validation.
- **Noise looks like outliers** → hard to separate.
- **Interpretability**: you must justify why something was flagged.
- Handling **high dimensionality** (distances lose meaning).

6.2 Categories of Anomaly Detection Schemes

Briefly explain different types of anomaly detection schemes. 4 6

81 Ba, 78 Bh 78 Bh asks to compare and contrast three methods.

(A) By how much labelling is available

- **Supervised**
 - Training set labelled both normal *and* outlier → treat as a **classification** problem.
 - Problem: **severe class imbalance** (outliers are rare) + the sample may not represent the outlier distribution.
 - Fixes: oversample outliers, make up artificial outliers, cost-sensitive learning.
- **Semi-supervised**
 - Only the **normal** class is labelled (common in practice).
 - Build a model of normality; anything not matching it is an outlier.
- **Unsupervised**
 - **No labels at all**. Assumes normal objects are *far more frequent* and cluster together.
 - Most practical, and the default assumption in exam answers.

(B) By the assumption made (this is the “three methods” answer for 78 Bh)

Method	Assumption	Weakness
Statistical (model-based)	Normal data follows a stochastic model ; low-probability objects are outliers.	Must guess the right distribution; poor in high dims.
Proximity-based	An outlier is far from its neighbours (or in a sparse region).	$O(n^2)$ cost; sensitive to r and k ; struggles w/ varying density (distance-based only).
Clustering-based	Normal objects belong to large, dense clusters; outliers do not.	Depends on the clustering; costly; outliers can distort the clusters themselves.

- **Classification-based** is sometimes given as a 4th: train a 1-class model of normality (eg 1-class SVM), flag whatever falls outside.

6.3 Distance / Proximity-Based Approaches

Explain distance-based methods for anomaly detection. 2+3 8 2+6 10 5 4+3

TOP 10/23 82 Bh, 81 Bh, 80 Ba, 79 Ba, 75 Ch, 75 Ash, 74 Ch, 73 Shr, 71 Ch, 70 Asa The single most repeated question in this chapter.

Core idea

- Judge an object by its **proximity to its neighbours**.
- An object is an outlier if its neighbourhood is **sparse** or its neighbours are **far**.
- Needs only a distance measure → Ch 2, §2.4, no distribution assumption.

1. Distance-based: the $DB(r, \pi)$ outlier

- Fix a **distance threshold** r (defines the neighbourhood) and a **fraction threshold** π .
- Object o is a $DB(r, \pi)$ -outlier if:

$$\frac{|\{o' \mid \text{dist}(o, o') \leq r\}|}{\|D\|} \leq \pi$$

- In words: **fewer than a π fraction of all objects lie within distance r of o** .
- **Equivalent k -NN form**: with $k = \lceil \pi \|D\| \rceil$, object o is an outlier if $\text{dist}(o, o_k) > r$, where o_k is its k th nearest neighbour.
- **Nested-loop algorithm**: for each object, scan the others and count how many fall within r ; stop early once the count exceeds πn . Cost $O(n^2)$.

2. k -NN score (the other common phrasing)

- Score each object by the **distance to its k th nearest neighbour** (or the average of its k nearest).
- Rank all objects by that score; the top ones are the outliers.
- 71 Shr asks exactly this, 10.

Clustering-based 76 Ash short note

- Cluster the data, then flag as outliers the objects that:
 - belong to **no cluster** (eg DBSCAN noise points),
 - are **far from their own cluster centroid**, or
 - belong to a cluster that is **very small or very sparse**.
- **DBSCAN** gives it free: **noise points** are exactly the outliers. → Ch 5
- + Unsupervised; reuses clustering you may already have.
- – Effectiveness depends entirely on the clustering; outliers can themselves distort the clusters.

3. Density-based: the local outlier 80 Bh 3

- **Why needed**: distance-based methods take a **global** view, using one r for the whole data set.
- If the data has **one dense cluster and one sparse cluster**, no single r works:
 - Set r small → the entire sparse cluster is wrongly flagged.
 - Set r large → genuine outliers beside the dense cluster are missed.
- **Fix**: compare an object's density to **the density of its own neighbours**, not to the whole data set.
- **Local Outlier Factor (LOF)**:
 - $LOF \approx 1$ → density like its neighbours → normal.
 - $LOF \gg 1$ → much sparser than its neighbours → outlier.
 - Built from *reachability distance* and *local reachability density*.
- **Distance-based finds global outliers; density-based finds local ones**. That contrast is the marks.

Strengths / weaknesses

- + No distribution assumption; simple; works on any data w/ a distance measure.
- – $O(n^2)$ in the naive form.
- – Sensitive to the choice of r, π, k .
- – Degrades in high dimensions (curse of dimensionality).

Statistical approaches

- Assume a model, flag low-probability objects.
- **Parametric**: assume a distribution (usually **Gaussian**), estimate its parameters, flag points in the tails.
 - Common rule: more than **3 standard deviations** from the mean.
 - **Grubbs' test**: detects one outlier at a time in univariate normal data, removes it, repeats. H_0 : no outlier.
- **Non-parametric**: no assumed shape; use histograms or kernel density estimates.
- **Convex hull method** 70 Ch asks this by name verified/added
 - Points on the **convex hull** (the boundary of the smallest convex polygon enclosing all points) are assumed to be **extreme**, hence outliers.
 - Peel the hull layer by layer to get successive outlier candidates.
 - **Limitation**: only ever finds *boundary* points. An outlier sitting *inside* the data cloud, in a sparse pocket, is never detected.

The likelihood (mixture-model) approach 8

70 Asa "What is anomaly detection? Explain likelihood approach for anomaly detection." Tan/Steinbach §10.2; not in Han & Kamber. verified/added

The model

- Treat the data set D as a **mixture of two distributions**:
 - M = the **majority** (normal) distribution, something tight such as a Gaussian.
 - A = the **anomaly** distribution, taken as **uniform** because nothing is assumed about how outliers are spread.
- λ = prior probability that an object is an anomaly, a small number fixed in advance.

$$D = (1 - \lambda) M + \lambda A$$

- At any moment the objects are partitioned into M_t and A_t . **Start with $M_0 = D$ and $A_0 = \emptyset$** : assume everything is normal.

Log-likelihood of the whole data set

$$LL_t(D) = |M_t| \log(1 - \lambda) + \sum_{x_i \in M_t} \log P_{M_t}(x_i) \\ + |A_t| \log \lambda + \sum_{x_i \in A_t} \log P_{A_t}(x_i)$$

- First two terms are the *normal* account, last two the *anomaly* account.

Algorithm

1. $M \leftarrow D$, $A \leftarrow \emptyset$, compute $LL_t(D)$.
2. For each point $x \in M$:
 - (a) Tentatively move x to A .
 - (b) Re-estimate M 's parameters without x and compute the new $LL_{t+1}(D)$.
 - (c) $\Delta = LL_{t+1}(D) - LL_t(D)$.
 - (d) If $\Delta > c$ (a threshold), **keep x in A and declare it an anomaly**; otherwise put it back in M .
3. Repeat until no point moves.

Why the test works (this is the explanation the marks are for)

- Moving a *normal* point out forces M to explain fewer points it already explained well, and the uniform A explains it badly $\rightarrow$ likelihood **falls**.
- Moving a *genuine outlier* out lets M tighten around the rest, and the outlier was costing M a huge penalty anyway $\rightarrow$ likelihood **rises sharply**.
- So **a large gain in likelihood is itself the evidence** that the point does not belong.

Strengths and weaknesses

- + Firm statistical footing; gives a principled test rather than an arbitrary distance.
- + Handles mixed / categorical attributes if a model exists for them.
- - Needs a distribution assumption for M , and a wrong one gives wrong outliers.
- - Expensive: M is re-estimated once per candidate point.
- - The threshold c and the prior λ are both guesses.
- - Degrades badly in **high dimensions**, like every parametric method.

6.4 Base Rate Fallacy

Base rate fallacy 3 5

70 Ch short note, 69 Ch 7, 76 Ch numerical **Numerical solved in the Ch 6 companion.** Not in Han & Kamber; from Tan/Steinbach and Axelsson.
verified/added

The fallacy

- **Base rate** = the prior probability of the rare event, eg $P(\text{intrusion})$.
- **Base rate fallacy** = the tendency to **ignore the base rate** and judge only by the test's accuracy.
- Consequence: when the event is very rare, **even a highly accurate detector produces mostly false alarms**, because the huge normal population generates more false positives than the tiny anomalous population generates true positives.

Set-up (intrusion detection)

- I = intrusive behaviour, $\neg I$ = normal. A = alarm, $\neg A$ = no alarm.
- **Detection rate** = $P(A|I)$ (true positive rate)
- **False alarm rate** = $P(A|\neg I)$
- **Bayesian detection rate** = $P(I|A) \rightarrow$ **what actually matters to the analyst: given an alarm, is it real?**

Bayes

$$P(I|A) = \frac{P(A|I)P(I)}{P(A|I)P(I) + P(A|\neg I)P(\neg I)}$$

Worked illustration

- Test 99% accurate both ways, disease prevalence 1 in 100.
- Then $P(\text{disease} | \text{positive})$ is only about **1/2**, not 99%.
- With a rarer event the result gets far worse.

Axelsson's conclusion

- In intrusion detection $P(I)$ is **tiny**, so the false alarm rate **dominates** the equation.
- $\rightarrow$ you need an **extremely low false alarm rate** to get a usable Bayesian detection rate. Raw accuracy is not enough.
- This is the formal reason **accuracy is the wrong metric** for anomaly detection $\rightarrow$ same argument as class imbalance in Ch 3, §3.7.

6.5 PYQ Mapping

N = solved in the Ch 6 Numerical companion. The syllabus gives this chapter no sub-topics, so PYQs are grouped by what they ask.

1. TOP 6/23 Anomaly detection + distance-based methods. The core question.

- "What is anomaly detection? Explain distance based method." 2+3 80 Ba
- same wording 8 73 Shr
- "Why is it important? + distance based method" 2+6 75 Ash
- "What do you mean by anomaly detection and why is it important? Describe distance based approaches." 4+3 74 Ch
- "Explain few distance based approaches" 10 71 Ch
- "What is outlier? Explain the distance base approaches." 5 75 Ch

2. HOT 3/23 Definition + importance + types of schemes.

- "What is it? Why important? Briefly explain different types of schemes." 2+2+4 81 Ba
- "Why is it important and where is it applicable?" 3+3 79 Bh
- "Discuss its importance in security." 5 72 Ka

3. PIN 1/23 Types of outlier w/ examples + how density-based detection works.

5+3 80 Bh

4. PIN 1/23 Compare and contrast 3 different methods of anomaly detection.

6 78 Bh

5. PIN 1/23 Issues related to anomaly detection 2 76 Ch; N false alarm rate / detection rate numerical

3 76 Ch

6. PIN 1/23 How can the Nearest-Neighbor algorithm be used for anomaly detection?

10 71 Shr

7. PIN 4/23 Short notes:

- Clustering and its application in anomaly detection 3 76 Ash
- Issues in anomaly / fraud detection 4 72 Ch
- Data Mining for Anomaly Detection 4 74 Ash
- **Anomaly Detection; Base Rate Fallacy; Convex Hull Method** 15 70 Ch

Papers added by the 2026-09 merge

The six papers of DM4.1BCT.pdf (2082 Bh, 2081 Bh, 2079 Ba, 2073 Ch, 2070 Asa, 2069 Ch), indexed here for the first time — the notes were built before that scan was merged. **N** = solved in this chapter's companion. verified/added

- What is anomaly detection? + distance-based method. 2+5 82 Bh
- Anomaly detection + the different **approaches**. 2+6 81 Bh
- **Strengths and weaknesses** of the statistical, proximity-based, density-based and cluster-based approaches. 6 79 Ba
- Anomaly detection + the **issues** associated with it. 2+3 73 Ch
- Anomaly detection + the **likelihood approach**. 8 70 Asa verified/added
- What is **Base Rate Fallacy**? Explain w/ example. 7 69 Ch

Coverage check

- All Ch 6 PYQs covered. 1 numerical 76 Ch → companion.
- **Gaps filled by research** verified/added :
 - **Convex hull method** 70 Ch: named in the Tan slides but never explained, and absent from Han & Kamber. Written out above w/ its limitation.
 - **Base rate fallacy**: the notes give the psychology definition but not the intrusion-detection form. Axelsson's argument added.
 - **Grubbs' test** and the supervised/semi-supervised/unsupervised split.

Sources used for this chapter

- **Han & Kamber 3rd ed**, ch 12 Outlier Detection: definition, the 3 outlier types, method taxonomy, $DB(r, \pi)$ formula, density-based / LOF. **Primary authority**.
- Anomaly Detection/Anomaly Detection.pdf (Tan, Steinbach & Kumar slides): convex hull, Grubbs' test, base rate fallacy, Axelsson.
- data mining_/Chapter 6 _ 7.pdf (Bhupesh K. Mishra), Intrusion.pptx, **DBK Lecture-17**.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
DM	data mining	DW	data warehouse
DB	database	DBMS	database management system
KDD	knowledge discovery in databases	DSS	decision support system
OLTP	on-line transaction processing	OLAP	on-line analytical processing
ROLAP	relational OLAP	MOLAP	multidimensional OLAP
HOLAP	hybrid OLAP	ETL	extract, transform, load
BI	business intelligence	ER	entity-relationship
attr(s)	attribute(s)	dim(s)	dimension(s)
multidim	multidimensional	txn(s)	transaction(s)
info	information	ops	operations
repo	repository	freq	frequent / frequency
config	configuration	no.	number
PCA	principal component analysis	SVD	singular value decomposition
SMC	simple matching coefficient	KNN	<i>k</i> -nearest neighbour
ANN	artificial neural network	POS	point of sale
FK	foreign key	UI	user interface
GIGO	garbage in, garbage out	w/, w/o	with, without
rm	remove	vs	versus
Eg	example / for example	+Fig	draw a figure
→	leads to / therefore	+	and

Exam-paper month codes (used in the year tags): Ba = Baishakh · Jth = Jestha · Asa = Ashar · Shr = Shrawan · Bh = Bhadra · Ash = Ashwin · Ka = Kartik · Mng = Mangsir · Po = Poush · Ma = Magh · Ch = Chaitra.

Tier chips: **TOP /23** = topic asked in 7 or more of the 17 papers · **HOT /23** = 4–6 papers · **PIN /23** = 1–3 papers but worth ≥ 8 marks. **Mark chips** such as **2+5** are the marks that question actually carried. verified/added marks anything researched and added that was not in the source notes.